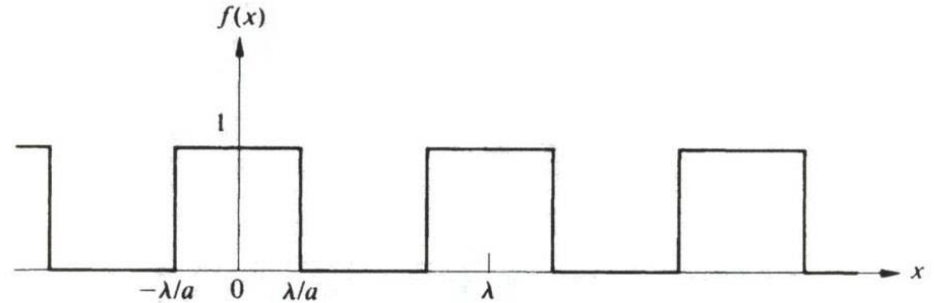


Example II: a square wave.

$$f(x) = \begin{cases} 1 & \text{when } -\frac{\lambda}{a} < x < \frac{\lambda}{a} \\ 0 & \text{others} \end{cases}$$



The Fourier series:

$$f(x) = \frac{2}{a} + \sum_{m=1}^{\infty} \frac{4}{a} \text{sinc}(m2\pi/a) \cos mkx$$

where $\text{sinc}(m2\pi/a) = \frac{\text{sinc}(m2\pi/a)}{m2\pi/a}$

The spatial period: λ ;

The spatial frequency: $\kappa=1/\lambda$;

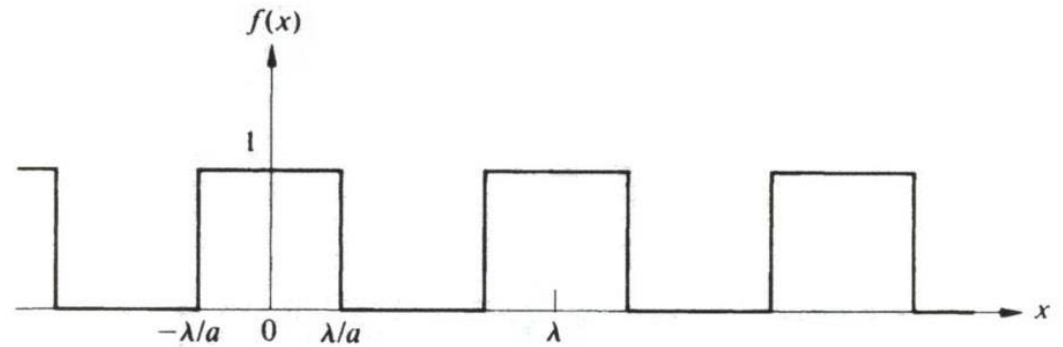
The angular spatial frequency: $k=2\pi\kappa$

Note: symbols k and λ for lightwave itself, $\mathbf{k}$ and $\boldsymbol{\lambda}$ to describe spatial optical patterns.

$$A_m = \frac{4}{a} \left(\frac{\sin m2\pi/a}{m2\pi/a} \right)$$

$$m = 1, 2, \dots$$

$$\text{So } A_1 = \frac{4}{a} \left(\frac{\sin 2\pi/a}{2\pi/a} \right)$$



$$\text{The ratio: } \frac{A_m}{A_1} = \frac{\sin m2\pi/a}{m\sin 2\pi/a}$$

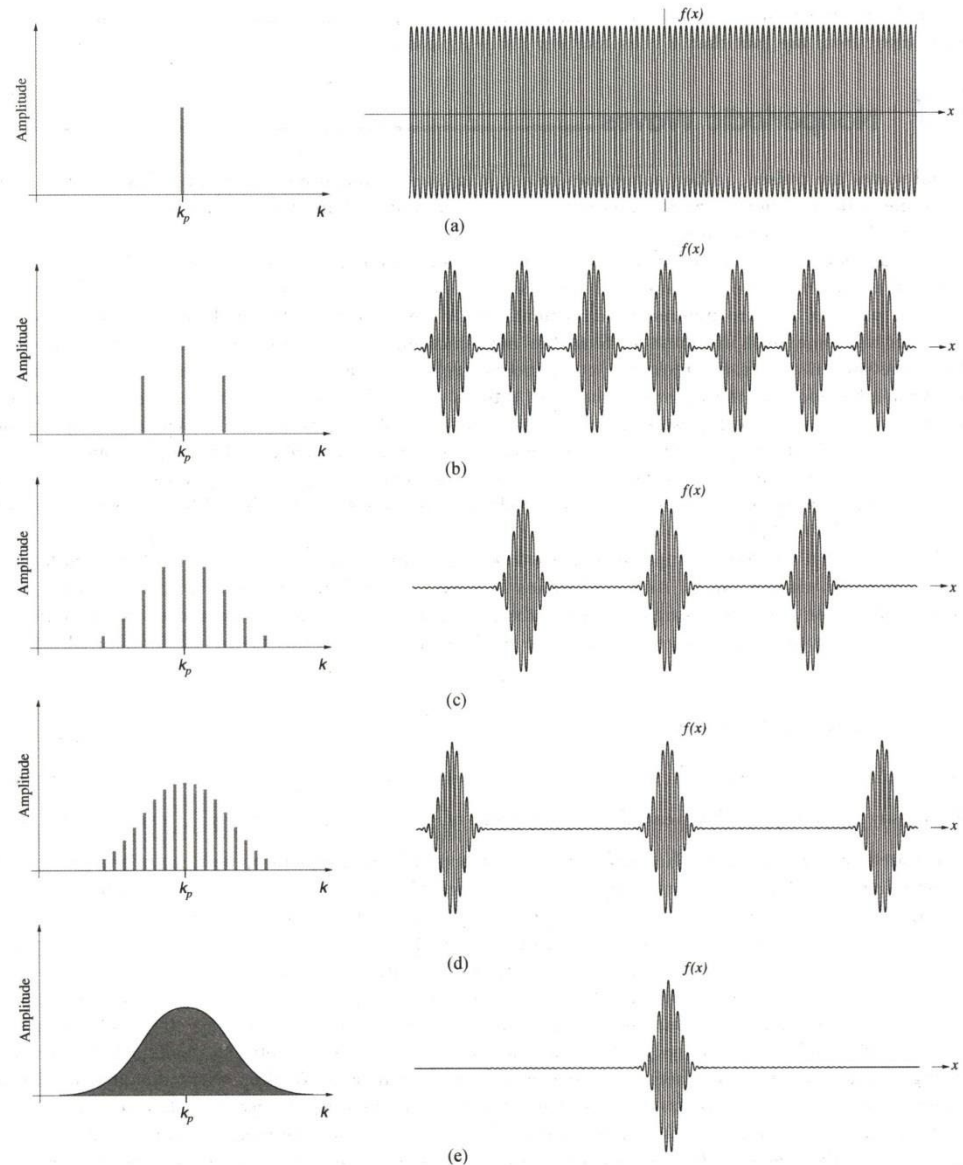
$$\text{For } a=4; m=9: A_9 \approx 10\% A_1$$

$$\text{For } a=400; m=9: A_9 \approx 99\% A_1$$

Making the peak narrower has the effect of introducing higher-order harmonics, which in turn have smaller wavelengths.

6.4 Nonperiodic Waves-Fourier Integrals

All real waves are pulses, albeit sometimes rather long ones, and so it is important to learn how to analyze nonperiodic functions.



Fourier integrals

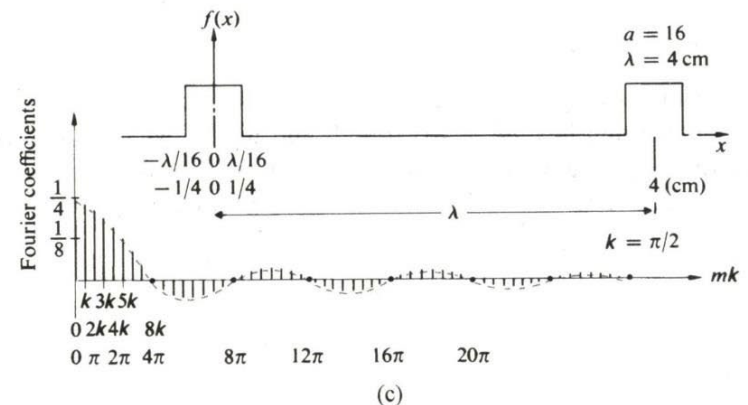
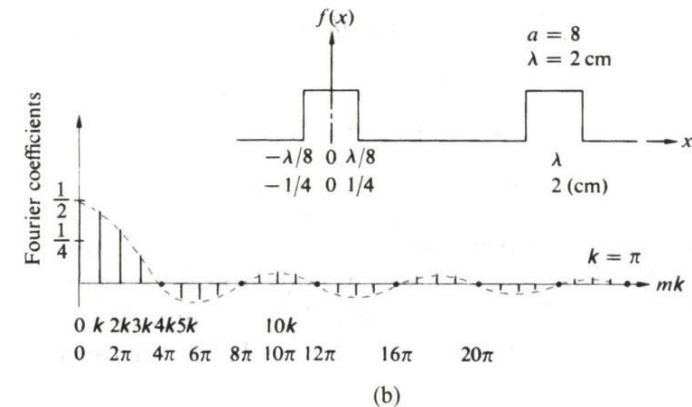
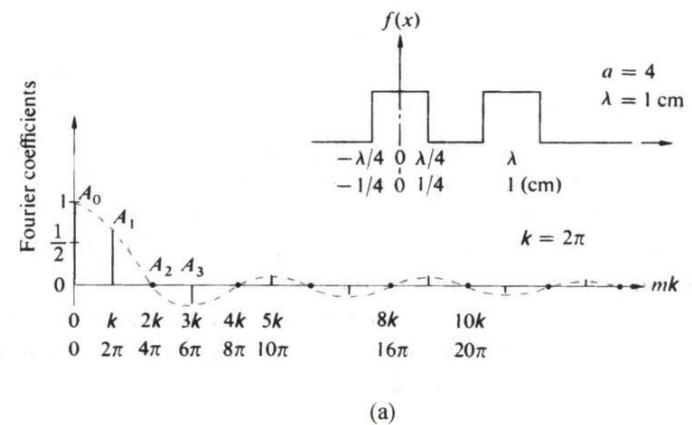
$$A_m = \frac{4}{a} \left(\frac{\sin m2\pi/a}{m2\pi/a} \right)$$

$$f(x) = \frac{2}{a} + \sum_{m=1}^{\infty} \frac{4}{a} \text{sinc } m2\pi/a \cos mkx$$

The importance of each particular frequency, mk , can be appreciated by examining the value of the corresponding Fourier coefficient, in this case A_m , which is thought of as the weighting factors that appropriately emphasize the various harmonics.

As λ increases and the function takes on the appearance of a single square pulse, the space between each of the $A(mk)$ contributions in the spectrum will decrease.

In the limit as λ approaches ∞ , the spectral lines will become infinitely close to each other.



As λ goes to infinity, Fourier integral changes from:

$$f(x) = \frac{A_0}{2} + \sum_{m=1}^{\infty} A_m \cos mkx + \sum_{m=1}^{\infty} B_m \sin mkx$$

$$A_0 = \frac{2}{\lambda} \int_0^{\lambda} f(x) dx \quad A_m = \frac{2}{\lambda} \int_0^{\lambda} f(x) \cos mkx dx \quad B_m = \frac{2}{\lambda} \int_0^{\lambda} f(x) \sin mkx dx$$

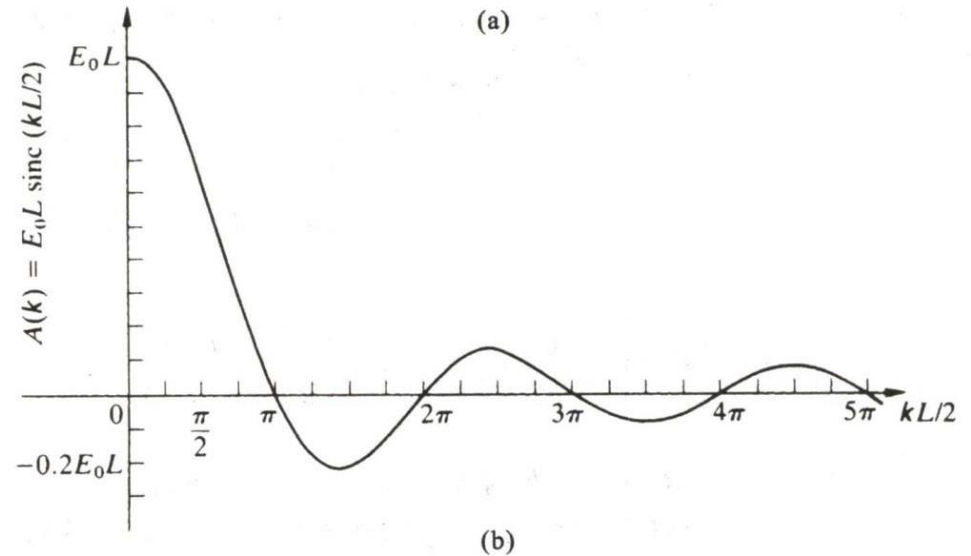
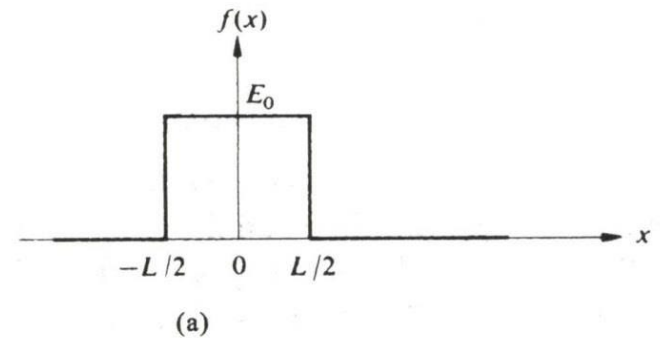
to

$$f(x) = \frac{1}{\pi} \left[\int_0^{\infty} A(k) \cos kx dk + \int_0^{\infty} B(k) \sin kx dk \right]$$

$$A(k) = \int_{-\infty}^{\infty} f(x) \cos kx dx \quad B(k) = \int_{-\infty}^{\infty} f(x) \sin kx dx$$

Pulses and Wave Packets

$$f(x) = \begin{cases} E_0 & \text{when } |x| < \frac{L}{2} \\ 0 & \text{when } |x| > \frac{L}{2} \end{cases}$$

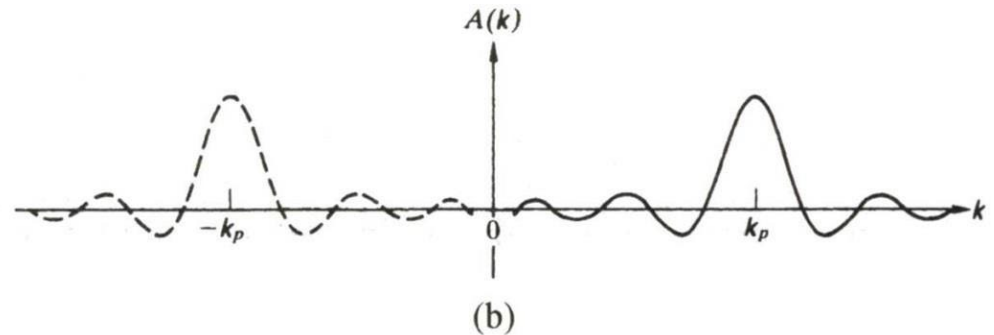
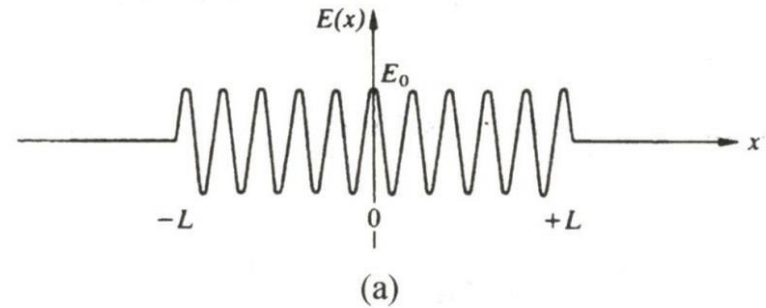


$$A(k) = E_0 L \left(\frac{\sin \frac{kL}{2}}{\frac{kL}{2}} \right) = E_0 L \operatorname{sinc} kL/2$$

The cosine wavetrain:

$$E(x) = \begin{cases} E_0 \cos k_p x & \text{when } -L \leq x \leq L \\ 0 & \text{when } |x| > L \end{cases}$$

k_p : the spatial frequency of the harmonic region of the pulse itself.



$$A(k) = E_0 L \left[\frac{\sin(k_p + k)L}{(k_p + k)L} + \frac{\sin(k_p - k)L}{(k_p - k)L} \right]$$

For the case of many waves in the train ($\lambda_p \ll L$)

$$k_p L \gg 2\pi \quad (k_p + k)L \gg 2\pi$$

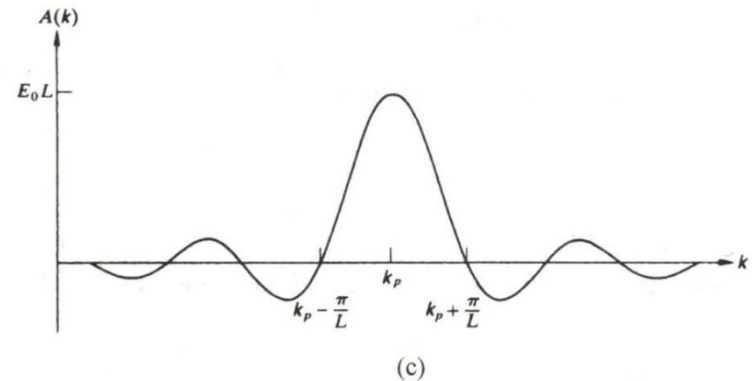
$\text{sinc}(k_p + k)L$ is a fairly small value.

Limit the treatment to the only positive values

$$k_p = k \quad \frac{\sin(k_p - k)L}{(k_p - k)L} = 1$$

$$A(k) = E_0 L \left[\frac{\sin(k_p + k)L}{(k_p + k)L} + \frac{\sin(k_p - k)L}{(k_p - k)L} \right]$$

$$A(k) = E_0 L \frac{\sin(k_p - k)L}{(k_p - k)L}$$



- ❑ It can be thought of as the composite of an infinite ensemble of harmonic waves, which is spoke **as wave packets** or **wave groups**.
- ❑ When $L \rightarrow \infty$, its frequency shrinks, and the curve closes down to a single tall spike at k_p (idealized monochromatic wave in the limit case).

The frequency bandwidths:

$$\Delta k = 2\pi/L$$

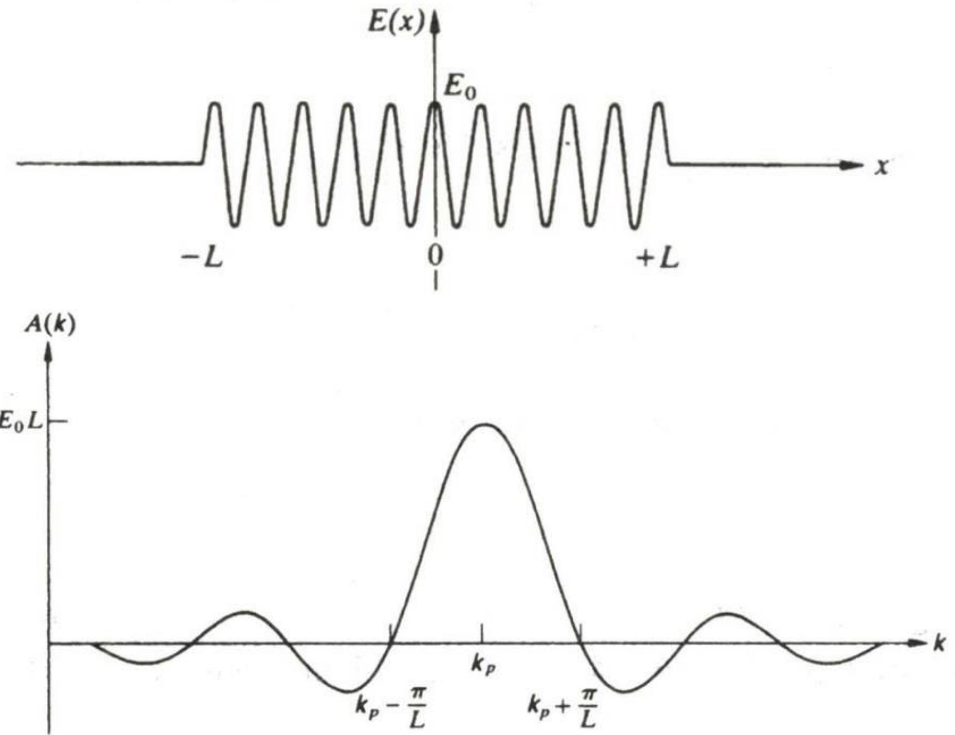
The spatial extent of the pulse:

$$\Delta x = 2L$$

The product:

$$\Delta k \Delta x = 4\pi$$

If the wave packet has a narrow bandwidth, it will extend over a large region of space and time.



The wave packet in the time domain:

$$E(t) = \begin{cases} E_0 \cos \omega_p t & \text{when } -T \leq t \leq T \\ 0 & \text{when } |t| > T \end{cases}$$

$$\Delta \omega = 2\pi/T$$

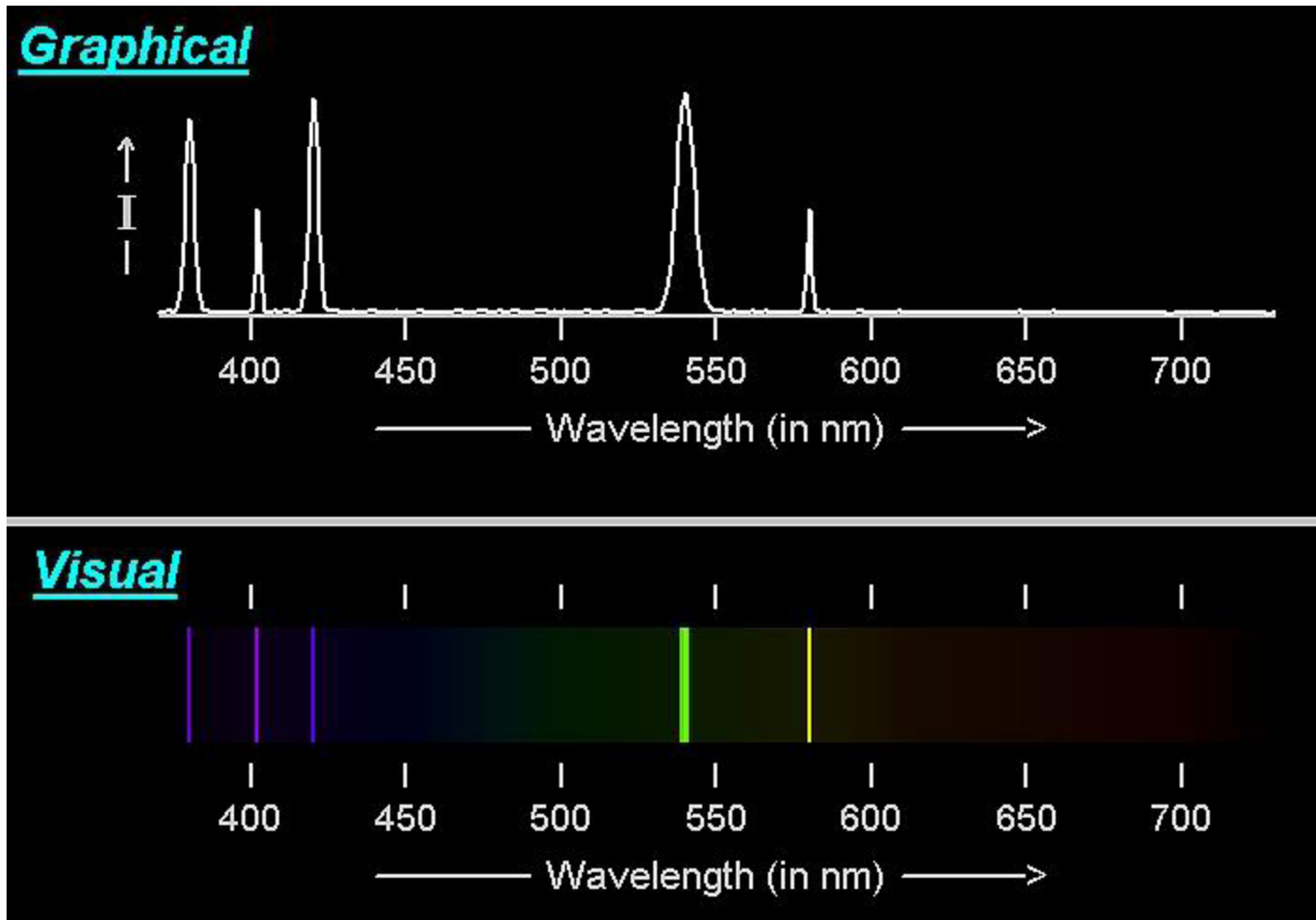
$$\Delta t = 2T$$

$$A(\omega) = E_0 T \operatorname{sinc}(\omega_p - \omega)T$$

$$\Delta t \Delta \omega = 4\pi$$

The k and ω are related by the phase velocity

Coherence Length



In any event, the individual spectral lines are never infinitely sharp, they always consist of a band of frequencies, however small.

Natural linewidth: The electron transitions responsible for the generation of light have a duration on the order of 10^{-8} or 10^{-9} s. Because the emitted wavetrains are finite, there will be a spread in the frequencies.

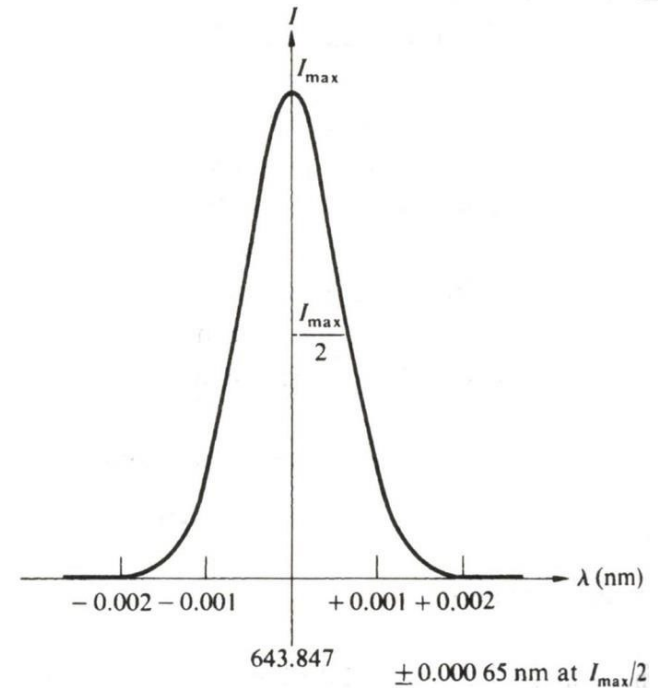
Other reason broadening the frequency distribution: **random thermal motion; collisions.**

Coherence time: $\Delta\nu \sim 1/\Delta t$

Coherence length: $\Delta x_c = c\Delta t_c$.

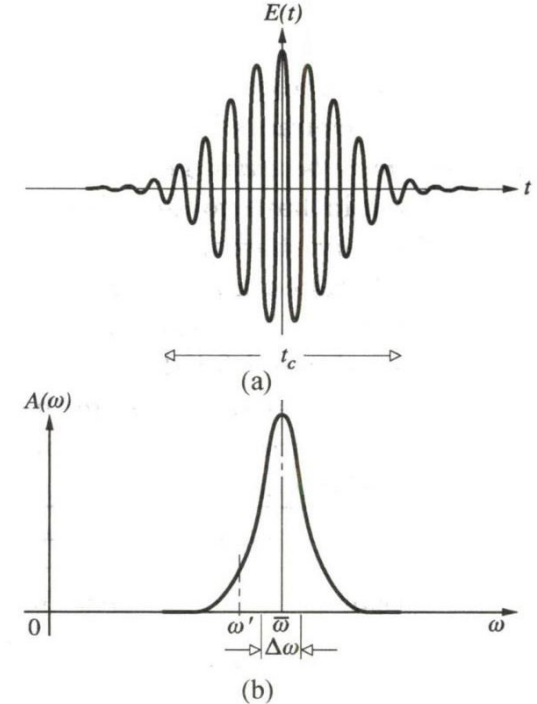
The coherence length is the extent in space over which the wave is nicely sinusoidal so that its phase can be predicted reliably.

Experimentally, the irradiance versus frequency is found to be Gaussian. So the net amplitude of the light field is also Gaussian.



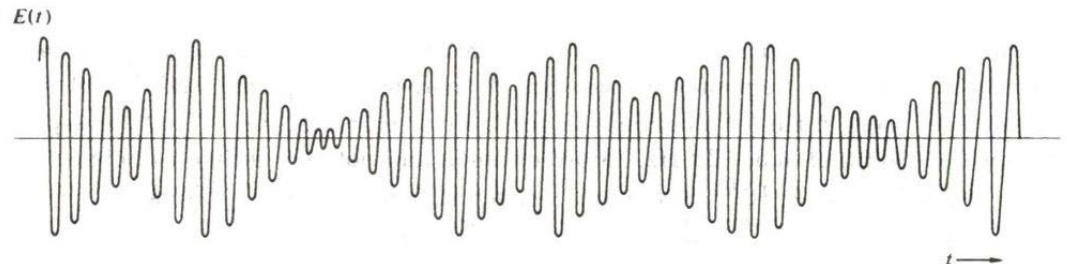
The frequency distribution of a spectral line from a quasimonochromatic source

- ❑ A single photon wavetrain: one of N identical such packets making up the beam.
- ❑ Consider the frequency component (ω'): an infinitely long, constant-amplitude sinusoid in a wavetrain; every packet has the identical amplitude and frequency.
- ❑ At any point of a stream of photons: the superposition of ω' component with random relative phase
- ❑ An average to a harmonic wave of frequency ω' with the amplitude proportional $N^{1/2}$.
- ❑ It is the same for every other frequency constituting the packets.



The observed spectral line (Gaussian function) corresponds to the power spectrum of the beam, it also corresponds to the power spectrum of an individual photon packet.

Different relative phases among wavetrains, so the composite lightwave can be pictured as following picture.



Homework

□ 7.32, 7.33, 7.37, 7.39

In Problem 7.33, the figure 7.20 should be 7.30